Data Analytics (DA)

Module-5-End

Assignment: Python For Data Analysis

Social Media Engagement Analytics Using Python

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INTRODUCTION

This project focuses on analysing social media engagement data to understand audience behaviour, content performance and engagement patterns. The dataset contains 5,000 records and 19 original columns, including user demographics, post types, content categories, likes, comments, shares, impressions, watch time, follower count, device type and sentiment.

OBJECTIVE

The main objective of this analysis is to identify the factors associated with higher social media engagement and generate meaningful insights that can support better content strategies.

DATA PREPARATION

The dataset was inspected and cleaned using Python and Pandas. Missing values were handled using appropriate techniques, data types were checked and converted where necessary, and duplicate records were examined.

Exploratory data analysis and statistical techniques were used to understand distributions, relationships and patterns. Visualizations were developed using Matplotlib, Seaborn and Plotly to present the findings effectively.

Tools Used

Python, Pandas, NumPy, Matplotlib, Seaborn, Plotly and Google Colab.

Final Insights and analysis

Content Performance

Post types have highest engagement

To identify which post type generate the highest average engagement, grouped the dataset by post type and calculated the average engagement rate for each category.

Output:

```
#Groupby Post_type and average engagement_rate
df.groupby("post_type")["engagement_rate"].mean().round(3)
```

post_type	engagement_rate
image	0.896
reel	0.783
text	1.065
video	1.122

dtype: float64

Video posts recorded the highest average engagement rate ie, 1.122, followed by the text post (1.065). Image post achieved an average of 0,896, while reels recorded the lowest average of 0.783.

This indicates that the video and text posts performed better in terms of engagement rate in this dataset, whereas reels had comparatively lower engagement.

Video is the best performing post type based on the average engagement rate.

Best-performing content category

To identify which content category generate the highest average engagement rate, grouped the dataset by content category and calculated average engagement rate. The results were sorted in descending order to identify the best-and lowest-performing categories.

Output:

```
#Best-performing content category
df.groupby("post_category")["engagement_rate"].mean().round(3).sort_values(ascending=False)
```

post_category	engagement_rate
food	1.359
tech	1.160
lifestyle	1.091
music	0.889
fitness	0.866
education	0.844
fashion	0.807
travel	0.702

dtype: float64

Food recorded the highest average engagement rate, followed by technology and lifestyle, whereas travel recorded the lowest. All other categories showed relatively similar engagement rates around 0.8.

Food is the best-performing content category, while travel is the lowest. Food content generated the highest average engagement, indicating stronger audience interaction with this category. Technology and lifestyle also performed well.

Countries have highest average engagement rate

To identify the countries having the highest engagement rate, grouped the dataset by country and calculated the average engagement rate for each country. The results were sorted in descending order to identify the best-and lowest-performing categories.

```
#country by average engagement rate
df.groupby("country")["engagement_rate"].mean().round(3).sort_values(ascending=False)
```

country	engagement_rate
Brazil	1.541
Australia	1.324
France	1.146
UAE	1.112
Canada	0.917
UK	0.851
Japan	0.770
Germany	0.759
India	0.655
USA	0.577

Brazil recorded the highest average engagement rate of 1.541, followed by Australia, France, and UAE whereas USA recorded the lowest.

The countries having the highest engagement rates are Brazil, Australia and France and UAE. The audience associated with Brazil appears the most responsive to the contents followed by Australia, France and UAE.

User Trends

How age affects engagement?

To understand how user age influence engagement with social media posts, grouped users by age and calculated the average engagement rate for each age group.

```
#How age affects engagement?
df.groupby("age")["engagement_rate"].mean().round(3).sort_values(ascending=False)
```

age	engagement_rate
24	2.897
55	2.599
51	2.397
48	1.676
50	1.553
41	1.540
35	1.522
47	1.402
42	1.351

Age 24 records the most engagement of 2.897, followed by 55 and 51, 2.599 and 2,397 respectively. The lowest recorded engagement rate of 0.385 belongs to the age group 28.

This clearly concludes that the engagement levels vary across the age groups, with some age groups showing the stronger interaction with social media content than others and out of all ages, 24 shows the most interaction.

Performance difference for verified accounts

To compare the average engagement rates of verified and non-verified accounts, grouped the dataset based on verification status, and calculated the average of the engagement rates.



```
#Performance difference for verified accounts
df.groupby("is_verified")["engagement_rate"].mean().round(3).sort_values(ascending=False)
```

is_verified	engagement_rate
True	1.054
False	0.955

dtype: float64

Verified accounts recorded a higher average engagement rate of 1.054 compared to non-verified profiles (0.955). This indicates that verified accounts received slightly stronger audience engagement on average in the dataset.

Behavioural Insights

Best time of day for impressions

To identify the best time for generating impressions. The posted_at column was examined to find the date and time information. The column was stored in datatype "datetime64[ns]" format; however, it contained only dates without specific time. Therefore, the best time of the day for generating impressions could not be determined.



Limitation:

A timestamp containing hours and minutes would be required to perform the time-of-the day analysis.

Device type impact on watch time

To identify which device type has the highest average watch time and understand how watch time varies across devices. The dataset was grouped by device-type and the average watch time in seconds was calculated.

```
#Device type impact on watch time
df.groupby("device_type")["watch_time_sec"].mean().round(3).sort_values(ascending=False)
```

device_type	watch_time_sec
mobile	4087.831
tablet	3979.736
desktop	3974.793

dtype: float64

Mobile users recorded the highest average watch time of 4087.831 seconds, followed by tablet users at 3979.736 seconds and desktop users at 3974.793 seconds.
This indicates that users accessing the content through mobile devices spent slightly more time watching it on average than users on tablets and desktops.
Mobile was the best-performing device type in terms of average watch time in this dataset.

Sentiment Analysis

Which sentiment performs best

To identify which sentiment category performs best based on average engagement rate. Grouped the dataset by sentiment. Calculated the average engagement rate for each sentiment category.

```
#Which sentiment performs best
df.groupby("sentiment")["engagement_rate"].mean().round(3).sort_values(ascending=False)
```

sentiment	engagement_rate
negative	1.015
neutral	0.974
positive	0.938

dtype: float64

Negative sentiment posts achieved the highest average engagement rate (1.015), followed by neutral (0.974) and positive posts (0.938).

This suggests that negative sentiment posts generate slightly higher engagement than neutral and positive posts in this dataset.

Behaviour of negative/neutral sentiment posts

To identify Behaviour of negative/neutral sentiment posts based on average engagement rate, grouped dataset by sentiment and calculated the average engagement rate. Filtered only *Negative* and *Neutral* posts, then used .describe() to show their statistical behaviour (count, mean, std, min, quartiles, max).

```
#Behavior of negative/neutral sentiment posts
df[df["sentiment"].isin(["negative","neutral"])].groupby("sentiment")["engagement_rate"].describe().round(3)
```

sentiment	count	mean	std	min	25%	50%	75%	max
negative	991.0	1.015	4.598	0.020	0.150	0.252	0.475	68.686
neutral	1575.0	0.974	5.308	0.007	0.147	0.253	0.529	121.932

The dataset reveals distinct performance patterns between negative and neutral sentiment posts:

- Count of negative Sentiment Posts 991.Mean Engagement Rate: 1.015. Standard deviation of 4.598 indicates relatively stable performance. Median engagement rate is 0.252(50%), with most posts clustering between 0.150 (25%) and 0.475 (75%). Maximum engagement rate reaches 68.686, showing occasional spikes but generally contained.
- Count of neutral Sentiment Posts 1575. Mean Engagement Rate: 0.974 (slightly lower than negative posts). Higher standard deviation of 5.308 suggests more volatile performance. Median engagement rate is 0.253(50%), with most posts ranging between 0.147 (25%) and 0.529 (75%). Maximum engagement rate is 121.932, highlighting rare but extreme peaks.

Negative sentiment posts deliver steadier engagement and slightly outperform neutral posts on average. Neutral posts, however, exhibit higher volatility and can sometimes achieve exceptional engagement spikes.

Key Findings

i. Post Type Performance: Video posts recorded the highest average engagement rate (1.122), followed by text (1.065), image (0.896) and reels (0.783).
ii. Content Category Performance: Food was the best-performing category (1.36), followed by Technology (1.16) and Lifestyle (1.091). Travel recorded the lowest average engagement rate.
iii. Country-wise Performance: Brazil recorded the highest average engagement rate among the countries analysed.
iv. Age and Engagement: Users aged 24 recorded the highest average engagement rate (2.897), followed by ages 55 (2.599) and 51 (2.397). Age 28 recorded the lowest rate (0.385).
v. Account Verification: Verified accounts recorded a higher average engagement rate (1.054) than non-verified accounts (0.955).
vi. Posting Time: The dataset contained posting dates but no specific time information. Therefore, the best time of day for impressions could not be determined.
vii. Device and Watch Time: Mobile users recorded the highest average watch time (4087.831 seconds), followed by tablet (3979.736 seconds) and desktop users (3974.793 seconds).
viii. Sentiment Analysis: Negative posts recorded the highest average engagement rate (1.015), followed by neutral posts (0.974) and positive posts (0.938). This suggests that negative and neutral posts generated comparatively higher engagement in this dataset.

Conclusion

The analysis indicates that engagement varies across post types, content categories, age groups, account verification status and device types. Video and food-related content showed stronger average engagement, while mobile users recorded the highest average watch time. Negative sentiment posts also received comparatively higher engagement rates than neutral and positive posts.

These findings can help content creators and social media managers understand audience preferences and make more informed content decisions.

Overall, this project demonstrates the use of Python-based data cleaning, exploratory data analysis, statistical analysis and visualization to transform raw social media data into meaningful insights.